

Department of Physics, Princeton University

Graduate Preliminary Examination
Part I

Thursday, August 27, 2020

9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

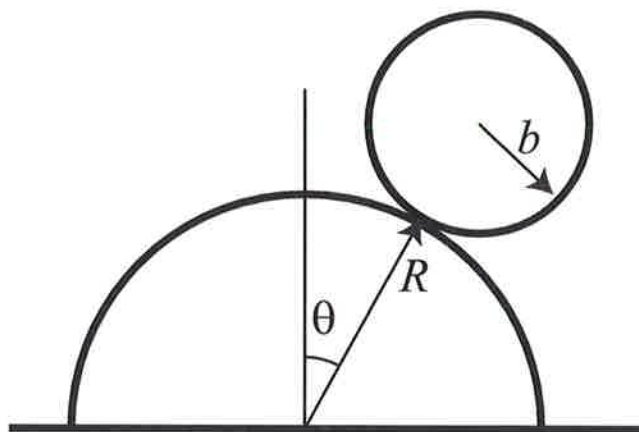
Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

1. Rolling cylinder

A uniform cylinder of mass m and radius b rolls off a fixed cylindrical surface of radius R under the influence of gravity. The axes of both cylinders are horizontal. The rolling cylinder starts from the top of the fixed cylinder with a negligibly small velocity.



- (a) If we assume the cylinder rolls without slipping, find the angle θ from the vertical when it loses contact with the fixed cylinder.
- (b) In practice for a finite value of friction coefficient μ the cylinder will start to slip before it loses contact. What value of μ would result in the cylinder starting to slip when $\theta = 45^\circ$?

2. Hanging Mass:

A point mass M is suspended by a vertical string of length L in the presence of gravity g . The upper end of the string is held fixed. The string is uniform and has a finite mass equal to m . You can assume that $M \gg m$, so that the tension in the string is constant. Consider small oscillations of the string and the mass in one vertical plane. Call the horizontal displacement of the string $y(z, t)$, where z is the distance below the point of support of the string (at $z = 0$), and t is time. In what follows, you may make approximations appropriate to the case of ‘small’ displacements.

- (a) What is the boundary condition on the string function $y(z, t)$ at the location of the mass M ?
- (b) Find the transcendental equation whose solutions give the frequencies of the normal modes of this system. Here only use $M \gg m$ to set the string tension constant, but otherwise give the exact equation for small amplitude oscillations.
- (c) Use the fact that $M \gg m$ to find approximate frequencies for the two lowest-frequency modes. Give explicit expressions for the associated time-dependent string motion for these two modes and give a simple physical interpretation of each motion.

3. Orbit stability

A particle of mass m moves under the influence of an attractive central force with potential $V(r)$.

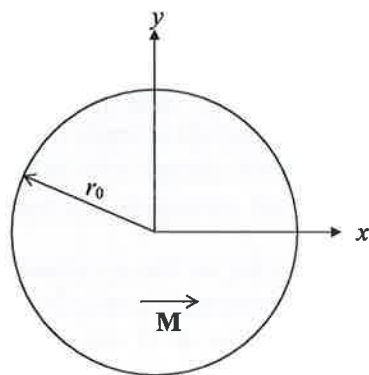
- (a) Suppose the particle moves in a circular orbit with angular momentum l . Derive the condition that gives the circular orbit radius r_l as a function of l and $V(r)$.
- (b) Now consider a small perturbation $\delta r(t)$ around such a circular orbit. What is the condition on the potential $V(r)$ such that the circular orbit remains stable?
- (c) Consider the special case $V(r) = -k/r^n$ with $n > 0$. For what values of n and l is the circular orbit stable?
- (d) Is it possible that the period of oscillations of the deviation from the circular orbit $\delta r(t)$ matches the period of the circular orbit? If it happens, what is the value of n and what does it imply about the long-term trajectory of the particle?

Section B. Electricity and Magnetism

1. Magnetic monopole detector

Some theories of elementary particles predict the existence of particles, called magnetic monopoles, that carry a magnetic charge g . The passage of a magnetic monopole through a loop of wire will induce an electric current in the loop that can be used as a detection signal. Note that magnetic charge acts as a source for the B field (in particular, $\vec{\nabla} \cdot \vec{B} = \mu_0 \rho_m$) so that a point monopole will have a Coulomb-like B field.

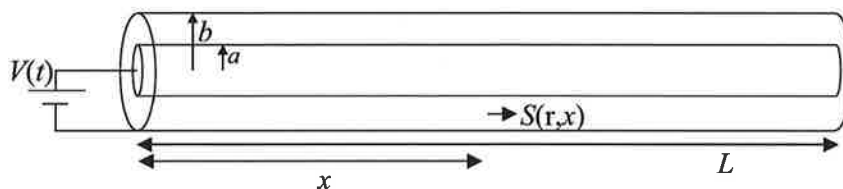
- (a) Consider a wire loop of radius b lying in the xy plane at $z = 0$ and centered on the z -axis. Consider a magnetic monopole g slowly moving up the z -axis from $z = -\infty$. Compute the magnetic flux Φ of the monopole B-field through the interior of the loop as a function of monopole position $z < 0$.
- (b) Let the loop have zero resistance, self-inductance L , and let there initially be no current in the loop. As the monopole moves, the changing flux through the loop of the monopole's B field leads to a Faraday EMF around the loop that causes a current I to flow. Find the current $I(t)$ if the monopole moves up the z -axis at constant speed $v \ll c$.
- (c) The induced current I creates an induced magnetic field at the location of the monopole, against which work must be done to keep the monopole moving at constant velocity. Show that the rate of doing mechanical work is exactly balanced by the the rate of increase of the inductive field energy $U = \frac{1}{2}LI^2$.

2. Moving Bar Magnet

- (a) A long cylinder of radius r_0 has uniform permanent magnetization density $\mathbf{M}$ perpendicular to the axis of the cylinder. Find the fields $\mathbf{B}$ and $\mathbf{H}$ everywhere. Let $\hat{z}$ = axis of the cylinder, and $\mathbf{M} = M\hat{x}$ as shown.
- (b) Suppose the cylinder is given uniform velocity $\mathbf{v} = v\hat{z}$ along its axis. Find the resulting charge density and electric field everywhere. You may ignore effects of order $(v/c)^2$.

3. Coaxial cable

A long co-axial cable is connected to a power supply with a variable voltage $V(t)$. The cable has an inner perfect conductor of radius a and an outer perfect conductor with inner radius b . The length of the cable L is much greater than b . The space between the conductors is empty.



a) The voltage $V(t)$ is very slowly ramped up from zero to a finite value V_0 (on a time scale much longer than L/c). There is nothing connected to the other end of the cable.

Find the Poynting vector $S(r, x)$ a distance x into the cable and show that the time integral of the energy flux at x is equal to the final electromagnetic energy stored from x to the end of the cable.

b) Now the voltage is kept constant at V_0 and a resistor R is connected to the other end of the cable, so a steady-state current is flowing through it.

Find linear momentum stored in the cable and explain how a system with no moving parts can have a finite linear momentum.

Department of Physics, Princeton University

Graduate Preliminary Examination
Part II

Friday, August 28, 2020

9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. Squeezing the Harmonic Oscillator

The quantum harmonic oscillator, described by the Hamiltonian $H_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$, with $[p, q] = \frac{\hbar}{i}$, is used to describe a remarkably diverse set of physical systems. Consider the class of harmonic oscillator ‘squeezed state’ wave functions of the form

$$\psi(q) = C \exp(-\alpha q^2/2)$$

where C is a normalization constant and α is an arbitrary *complex* number (with positive real part to ensure that the wave function is normalizable). States of this type can be prepared from the ground state by turning on a suitable interaction hamiltonian for the right length of time. It is not in general an energy eigenstate, but it has interesting and useful properties:

(a) The means of p and q in this state vanish. Calculate their variances and show that for *real* α this is a minimum uncertainty state, i.e. a state with $(\delta p)(\delta q) = \hbar/2$ (define $(\delta q)^2 = \langle (q - \langle q \rangle)^2 \rangle$, and similarly for δp).

(b) This state satisfies the equation

$$(\alpha q + ip/\hbar)\psi(q) = (\alpha q + \frac{\partial}{\partial q})\psi(q) = 0$$

Show that the time-evolved state $\psi_t(q) = \exp(-iH_0 t/\hbar)\psi(q)$ satisfies the same equation with a time-dependent (and in general complex) $\alpha(t)$. One way to do this is to go to the Heisenberg picture and time-evolve the operators p and q , rather than $\psi(q)$.

(c) You should find that $\alpha(t)$ becomes complex with time, even if $\alpha(0) = \alpha$ is real. So, in general, at $t > 0$ the state is no longer minimum uncertainty. Show that there are later times in the harmonic oscillator period where $\alpha(t)$ becomes real again (and the state recovers its minimum uncertainty character). Identify these real values of $\alpha(t)$ and comment on the nature of the minimum uncertainty states that are visited (assume that α is large, so that the initial minimum uncertainty state has small variance in q (and large variance in p)).

2. Low-energy scattering

Consider the problem of s-wave ($l = 0$) scattering of a particle of mass m from an attractive square-well potential of depth V_0 and radius r_0 : $V(r) = -V_0\theta(r_0 - r)$ in three dimensions.

- a) First consider the problem of s-wave bound states in this potential. Show that there is a critical potential strength V_{crit} such that for $0 < V_0 < V_{crit}$ there are no bound states. To put it another way, show that a bound state first appears when $V_0 = V_{crit}$. Determine the value of V_{crit} .
- b) Now consider scattering of a particle of momentum k from this potential. Set up the equation for determining the phase shift $\delta_0(k)$ and show that it implies that $\delta_0 \sim Ak$ as $k \rightarrow 0$. Evaluate the coefficient A as a function of V_0 .
- c) Calculate the contribution of the s-wave phase shift to the total cross section in the limit of small k . How does the zero-energy cross section behave: (i) in the limit $V_0 \rightarrow 0$?; (ii) in the limit $V_0 \rightarrow V_{crit}$? Comment and explain.

3. Hydrogen transition

A hydrogen atom located at rest at $\vec{r} = 0$ is initially in the 2s state when an ion of charge Q passes by it. Assume the ion moves with a constant velocity $\vec{v} = v\hat{y}$ on a straight line whose closest approach to the hydrogen atom is $\vec{b} = b\hat{z}$, with $b \gg a_B$, where a_B is the Bohr radius. While the ion passes by, the electron in the hydrogen atom experiences a time-dependent potential

$$V_1(\vec{r}, t) = \frac{Qe}{|\vec{b} + \vec{v}t - \vec{r}|}.$$

We are interested in calculating the transition probability of the hydrogen atom to one of the 2p states. In this problem you may assume that 2s and 2p states are degenerate.

- Find an expression of $V_1(\vec{r}, t)$ that is valid at all times and is appropriate for calculating the transition probability in the limit $b \gg a_B$. Identify the leading term in this expansion that will give a non-vanishing transition amplitude between 2s and at least one of the 2p states in first-order time-dependent perturbation theory.
- Using first-order time-dependent perturbation theory, calculate to leading order in a_B/b the probability that the atom winds up in a 2p state.

Some useful hydrogen atom wave-functions are:

$$\begin{aligned}\psi_{2s} &= \frac{1}{2\sqrt{2\pi a_B^3}} \left(1 - \frac{r}{2a_B}\right) e^{-r/(2a_B)} \\ \psi_{2p,0} &= \frac{z}{4\sqrt{2\pi a_B^5}} e^{-r/(2a_B)} \\ \psi_{2p,\pm 1} &= \frac{x \pm iy}{8\sqrt{\pi a_B^5}} e^{-r/(2a_B)}\end{aligned}$$

You may also find the following integral useful:

$$\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{3/2}} = 2$$

Section B. Statistical Mechanics and Thermodynamics

1. Relativistic electrons

Consider a high-density gas of electrons, as in a cold dense star. The number density of the electrons is n electrons per unit volume.

- (a) At high enough density n , essentially all of these electrons are moving relativistically, even in the ground state. How high does the density have to be for the ground state of this gas to be in this relativistic regime? Give your answer first as a formula in terms of physical constants, and then an order of magnitude number in electrons per cubic meter.
- (b) Calculate (i.e, do an integral) the ground state energy density u_0 of this gas as a function of the number density n and physical constants, assuming that we are fully in the relativistic regime, so the mass of the electron can be neglected. Also ignore any interactions between the electrons and between the electrons and the nucleons that are also present in the star.

2. Hawking radiation

Hawking and Bekenstein famously proposed that a black hole is in fact a thermodynamic system with a thermodynamic energy equal to its mass energy (Mc^2) and also endowed with a temperature $T(M)$ and an entropy $S(M)$. The mass can be equivalently expressed in terms of the Schwarzschild radius $R = 2GM/c^2$ (G is Newton's constant) of the black hole's spherical event horizon. Hawking further established that the entropy equals exactly one-quarter the area of the horizon (measured in Planck units), a relation that can be written as

$$S/k_B = \frac{1}{4} \frac{4\pi R^2}{\hbar G/c} = \frac{4\pi G M^2}{\hbar c} \quad (1)$$

In what follows, we will work out some consequences of the thermodynamic nature of the black hole.

- a) In thermodynamics, the temperature of a system is related to the derivative of the entropy with respect to the energy. Use this basic relation to calculate the temperature of the black hole event horizon for a black hole of mass M .
- b) Assuming that the black hole only emits photons with a black-body radiation spectrum from the surface of the horizon which is at this temperature, find the rate of energy loss. Express your answer in terms of fundamental constants. You might find useful the integral $\int_0^\infty x^3 dx / (e^x - 1) = \pi^4/15$.
- c) Find the time it takes a black hole of mass M to evaporate due to this energy (thus mass) loss.

3. Brownian Motion

A solid spherical particle of radius b and mass M is suspended in a fluid, and as seen using an optical microscope, undergoes Brownian motion with trajectory $\vec{r}(t)$. In this problem you are asked to show that a measurement of the mean-square displacement $\langle |\vec{r}(t_1) - \vec{r}(t_2)|^2 \rangle$ can be used to determine the Boltzmann's constant k_B .

Assume the densities of the solid and the fluid are identical, so buoyancy can be ignored. The cause of the Brownian motion is the rapidly fluctuating random force $\vec{F}(t)$ that is due to collisions with the molecules in the fluid. The force has zero mean, $\langle \vec{F}(t) \rangle = 0$, and its components (F_x, F_y, F_z) have the following two-time correlation function:

$$\langle F_\alpha(t_1) F_\beta(t_2) \rangle = C \delta_{\alpha,\beta} \delta(t_1 - t_2)$$

in terms of the Kronecker and Dirac delta functions. The fluid has viscosity η and the system is in equilibrium at a temperature T . The equation of motion of the particle can be written as

$$M \frac{d^2 \vec{r}(t)}{dt^2} + 6\pi\eta b \frac{d\vec{r}(t)}{dt} = \vec{F}(t).$$

- (a) For a single realization of these random forces, express the velocity $\vec{v}(t) = \frac{d\vec{r}(t)}{dt}$ as an integral involving the past forces, $\vec{F}(t')$ for $t' < t$.
- (b) Find the coefficient C as a function of temperature T and other constants mentioned above.
- (c) Calculate the mean square displacement $\langle |\vec{r}(t_1) - \vec{r}(t_2)|^2 \rangle$ as a function of the time difference for large time differences $|t_2 - t_1|$, and explain how measuring this quantity can be used to determine Boltzmann's constant k_B , assuming you also have measurements of all other constants above, except k_B and C .

